Layered AFE Helmet Plaque - MSgt Renholm



Stack (back to front)

Layer 1 Backer
Layer 2 Field panel
Layer 3 Banner shadow outline, Helmet shadow outline
Layer 4 Banner tail L, Banner tail R, Far-side rim, Hose, Shell
Layer 5 Banner, Edge roll trim, Far-side liner, Hidden spacer, Hose ring, Mask seal, Shell dome, Visor lens edge
Layer 6 Bayonet receiver, Ear cup, Hose connector, Hose end fitting, Mask shell, Shell crown, Visor housing rim
Layer 7 Bayonet arm, Ear cup cover, Exhalation valve, Mic boss, Visor housing
Layer 8 Receiver screw, Valve cover, Visor end cap, Visor lock knob

Stain colours

A Dark Walnut
B Natural + clear coat
C Golden Pecan / Early American

45 front pieces + 6 back pieces
6 sheets of 3 mm basswood, 300 x 300 mm
Finished size 286 x 286 mm, ~24 mm deep

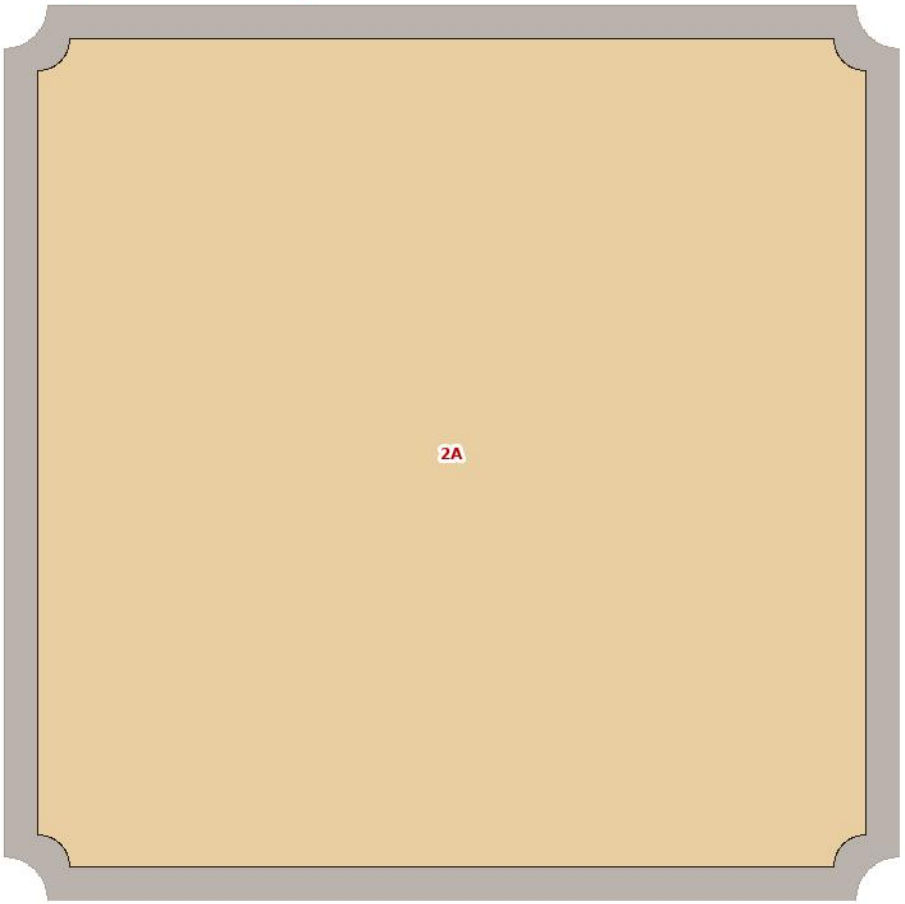
xTool S1 starting points for 3 mm basswood plywood (always run a material test first)

Colour	Operation	S1 20 W	S1 40 W
Black fill	Engrave (text, border)	30% / 200 mm/s / 0.08 mm	18% / 300 mm/s / 0.08 mm
Blue line	Score (glue guides)	12% / 120 mm/s / 1 pass	6% / 150 mm/s / 1 pass
Red line	Cut	100% / 8-10 mm/s / 1 pass	100% / 18-22 mm/s / 1 pass

Air assist ON for cutting. Engrave and score run before the cut (XCS default). Masking the top face keeps smoke stains off.

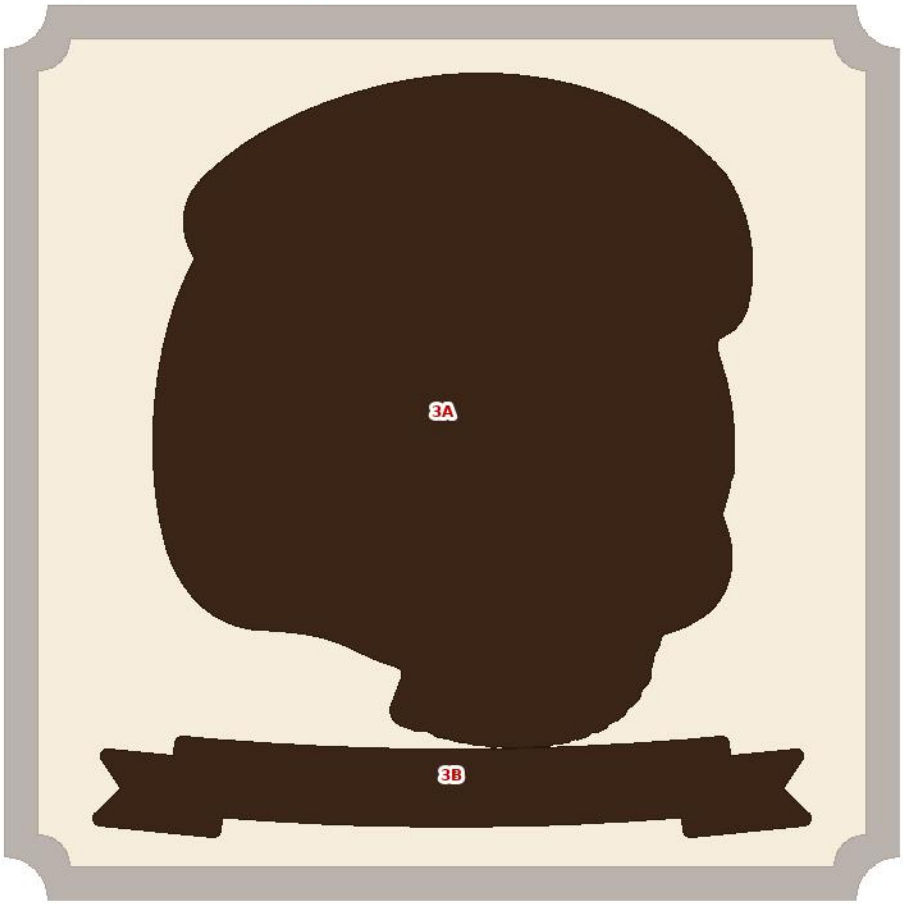
Glue-up - follow the scored outlines on the layer below

Step 1: add layer 2



Pieces: 2A Field panel

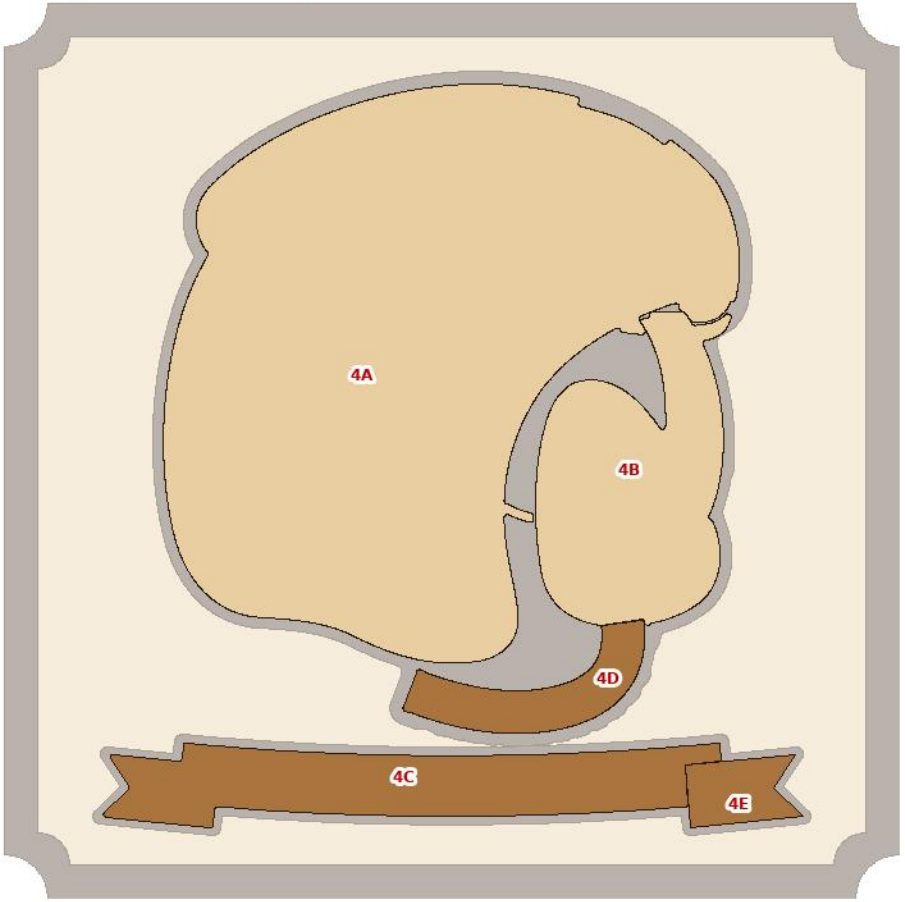
Step 2: add layer 3



Pieces: 3A Helmet shadow outline, 3B Banner shadow outline

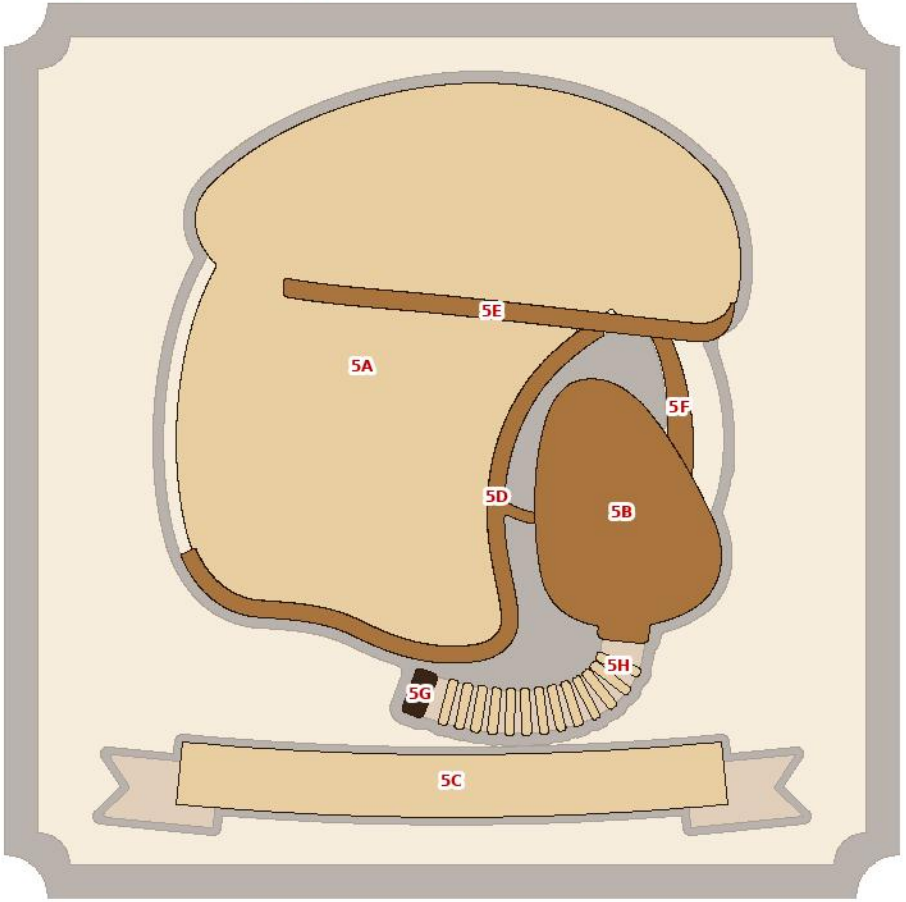
Glue-up - follow the scored outlines on the layer below

Step 3: add layer 4



Pieces: 4A Shell, 4B Far-side rim, 4C Banner tail L, 4D Hose, 4E Banner tail R

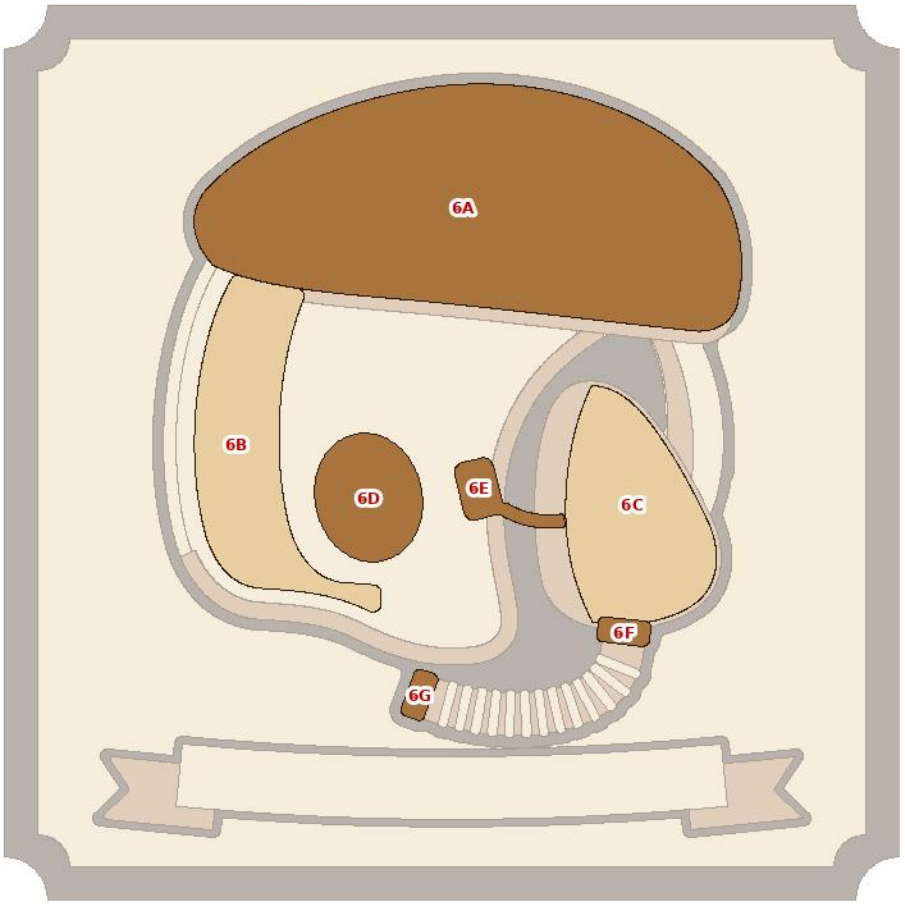
Step 4: add layer 5



Pieces: 5A Shell dome, 5B Mask seal, 5C Banner, 5D Edge roll trim, 5E Visor lens edge, 5F Far-side liner, 5G Hidden spacer, 5H Hose ring x13

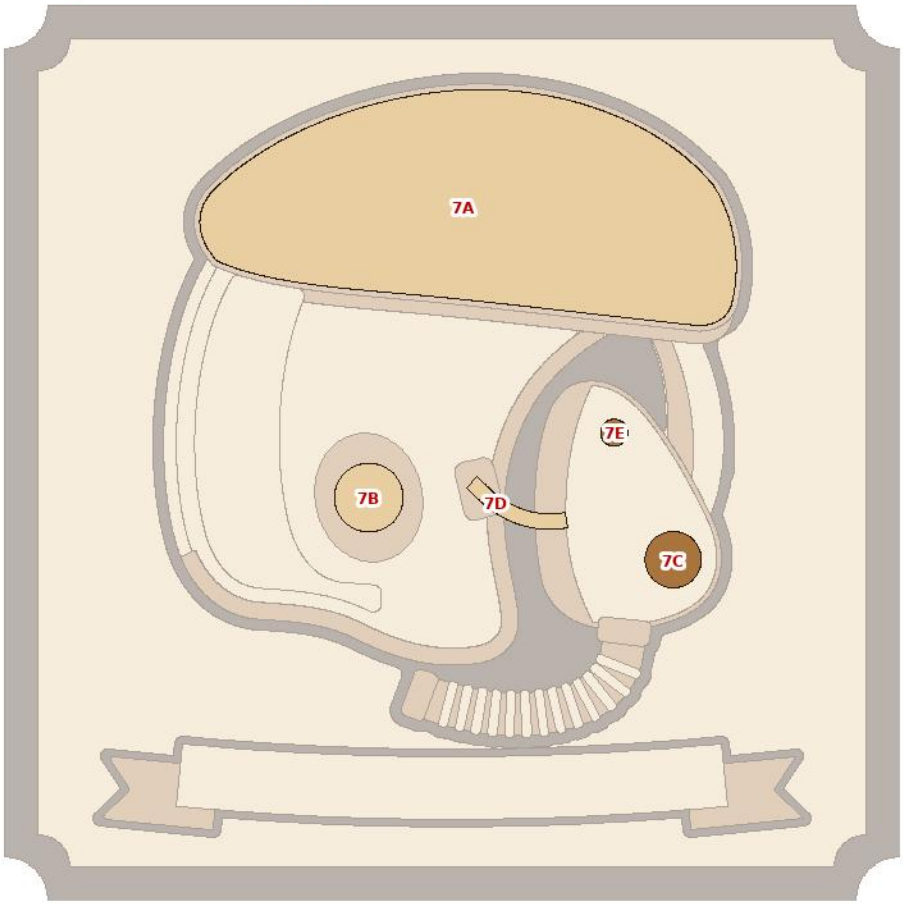
Glue-up - follow the scored outlines on the layer below

Step 5: add layer 6



Pieces: 6A Visor housing rim, 6B Shell crown, 6C Mask shell, 6D Ear cup, 6E Bayonet receiver, 6F Hose connector, 6G Hose end fitting

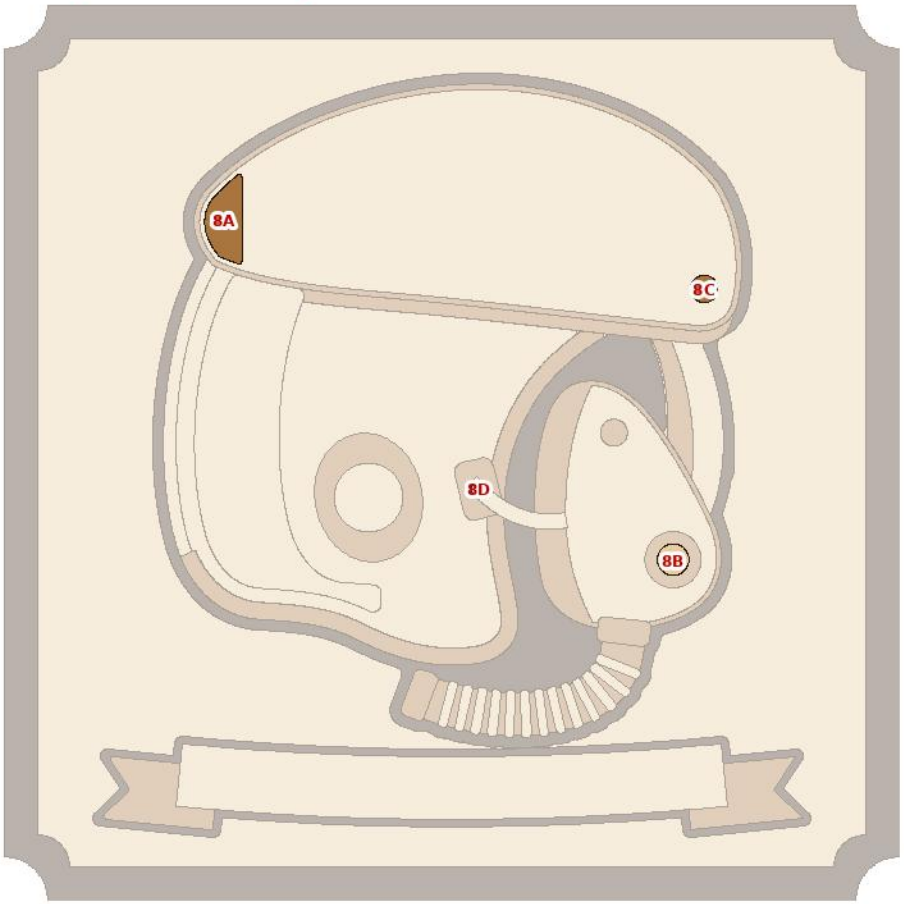
Step 6: add layer 7



Pieces: 7A Visor housing, 7B Ear cup cover, 7C Exhalation valve, 7D Bayonet arm, 7E Mic boss

Glue-up - follow the scored outlines on the layer below

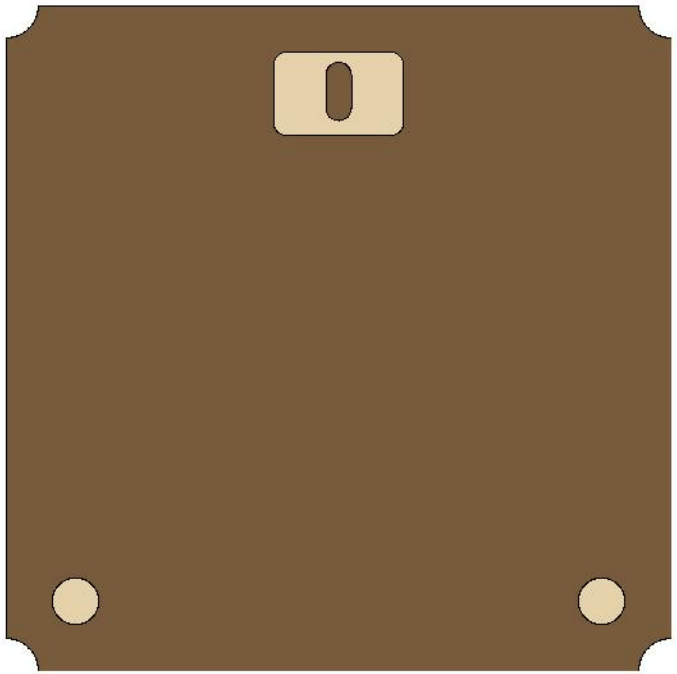
Step 7: add layer 8



Pieces: 8A Visor end cap, 8B Valve cover, 8C Visor lock knob, 8D Receiver screw

Back side - keyhole hanger and wall bumpers

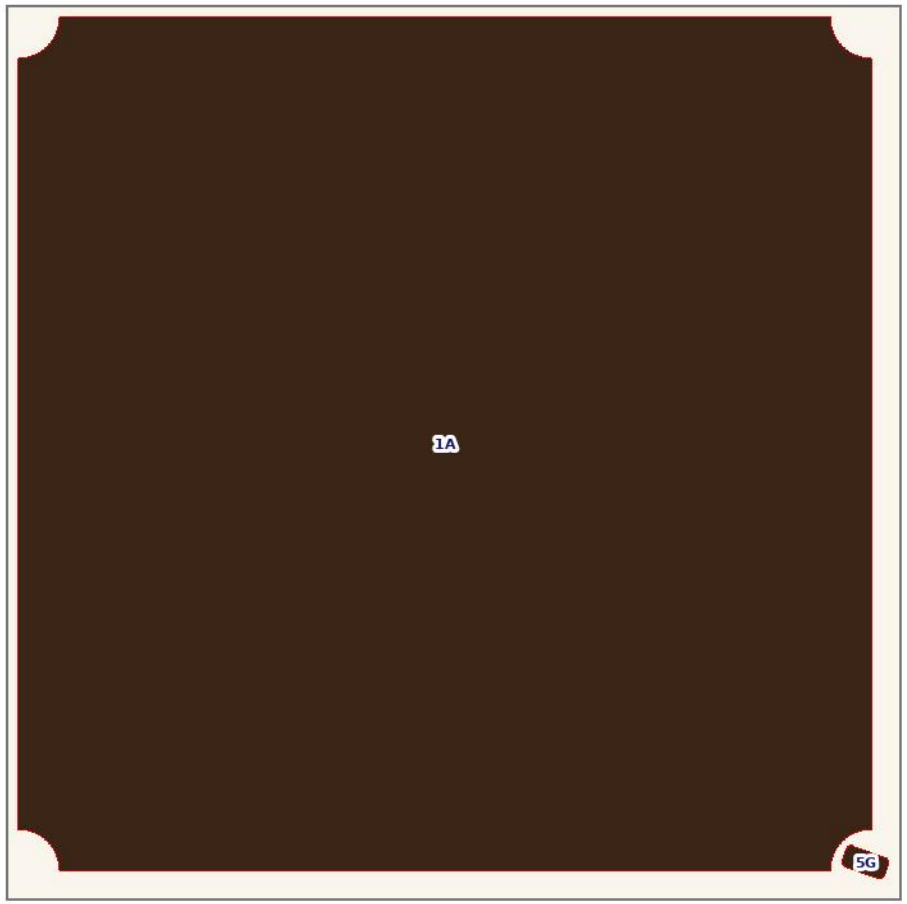
1. Glue the keyhole SPACER (the piece with the long rounded pocket) to the back of the backer, centred left-to-right, with its top edge ~20 mm below the top edge.
2. Glue the keyhole PLATE on top of it, keyhole lined up with the pocket, wide hole at the bottom.
3. Stack the four bumper discs into two pairs and glue one pair in each bottom corner (~20 mm in from the edges) so the plaque hangs flat.
4. Hang on a #8 pan-head screw left ~6 mm proud of the wall.



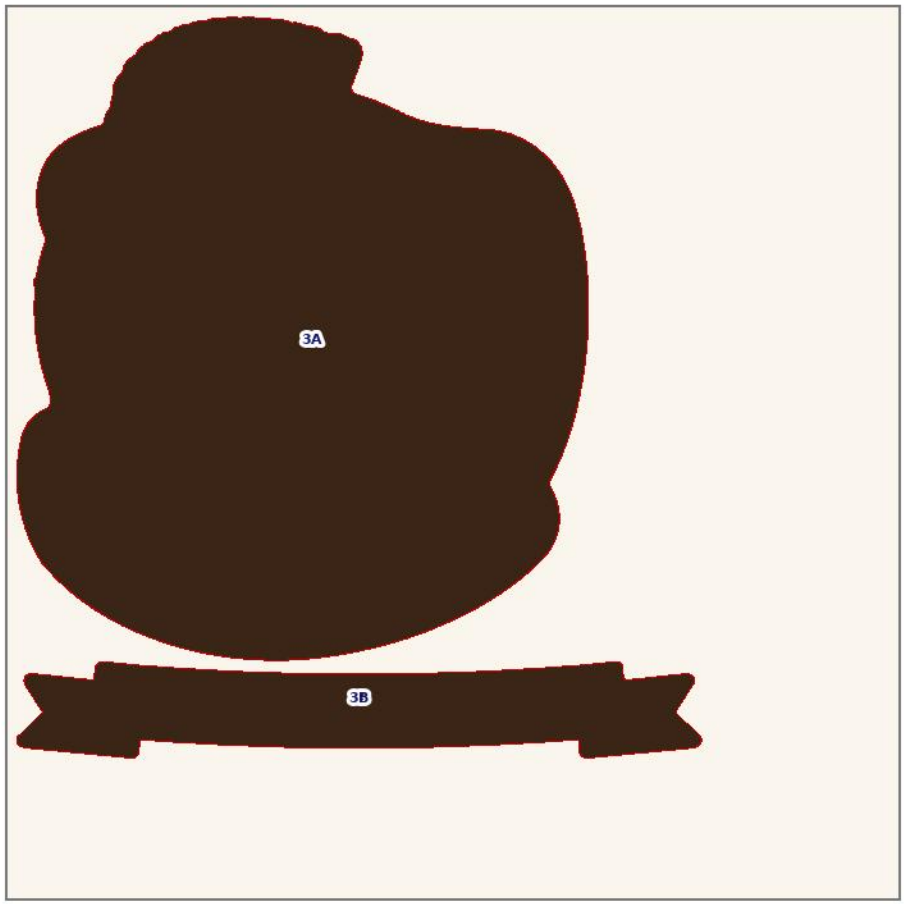
View from the BACK

Cut sheets - part IDs (larger pieces also have it engraved on a hidden spo

Sheet 1 - stain: Dark Walnut

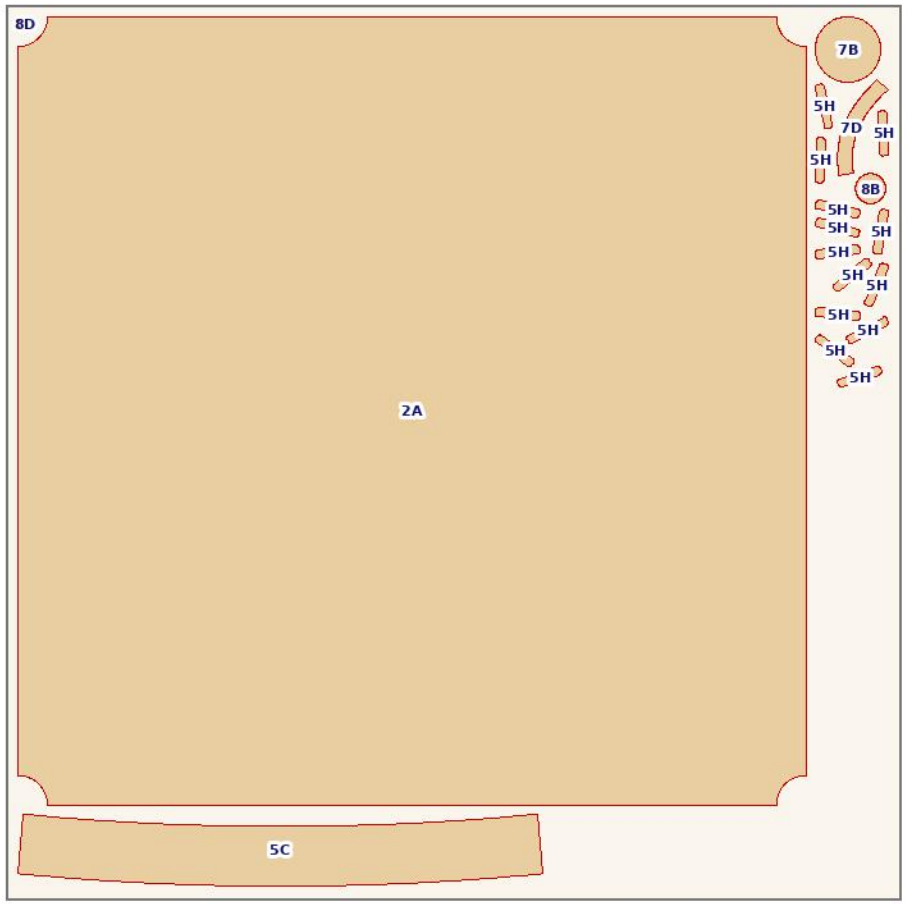


Sheet 2 - stain: Dark Walnut

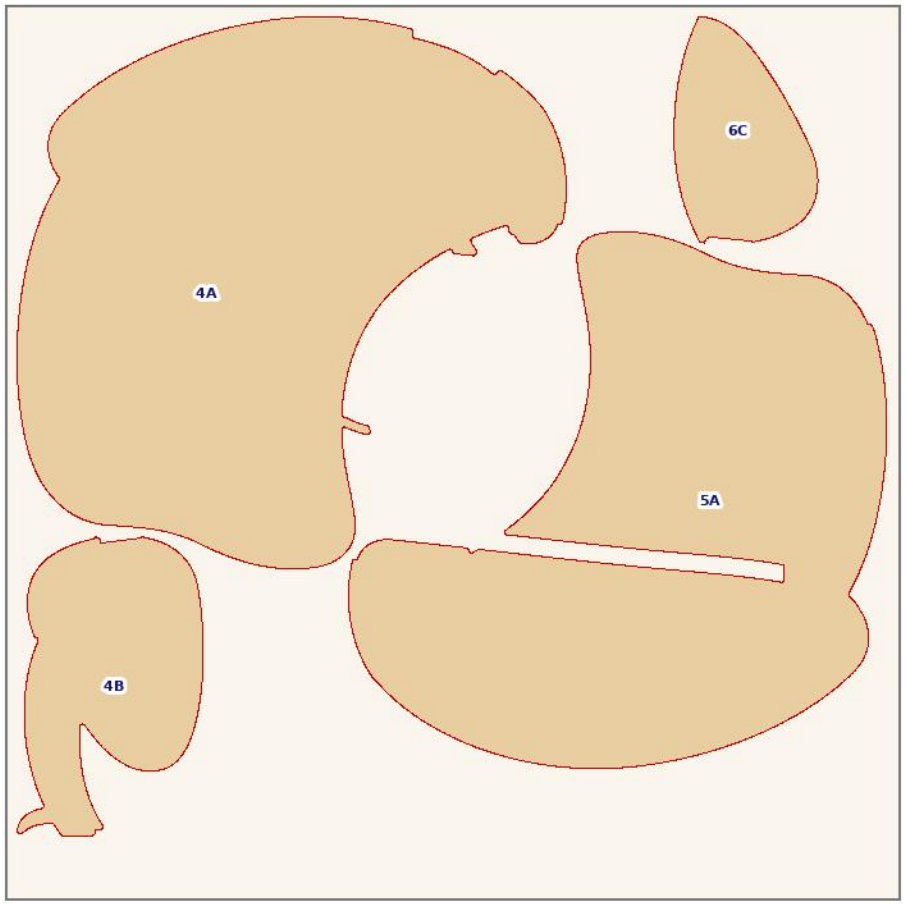


Cut sheets - part IDs (larger pieces also have it engraved on a hidden spo

Sheet 3 - stain: Natural + clear coat

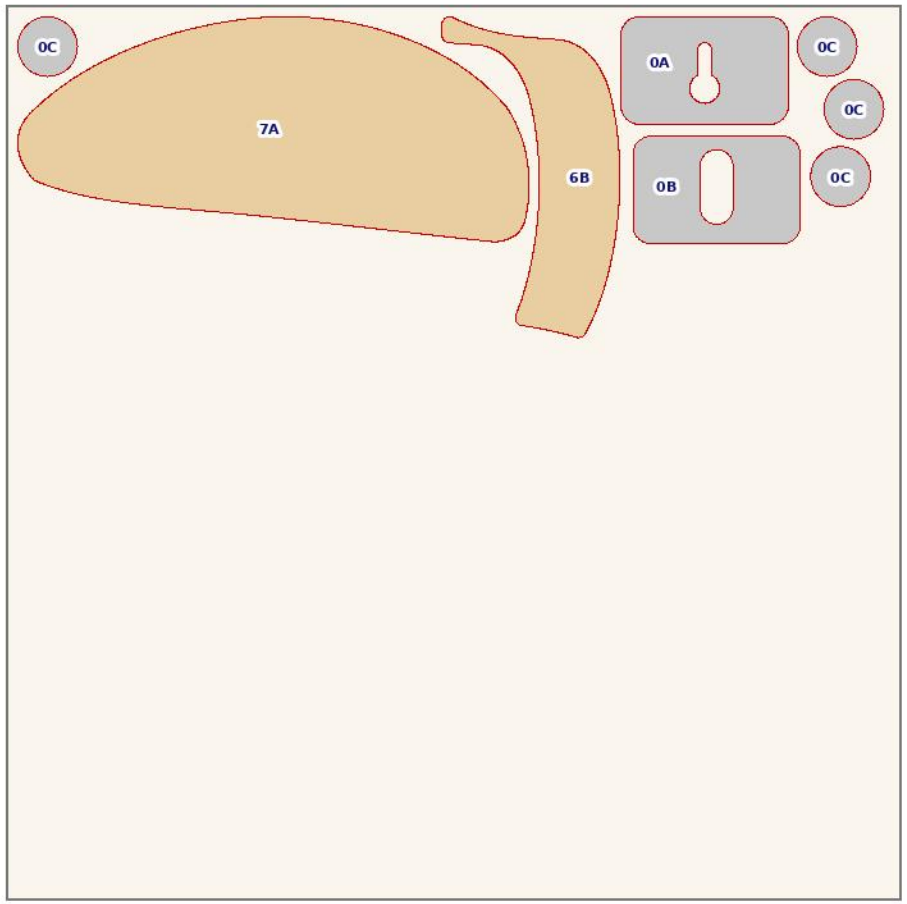


Sheet 4 - stain: Natural + clear coat



Cut sheets - part IDs (larger pieces also have it engraved on a hidden spo

Sheet 5 - stain: Natural + clear coat



Sheet 6 - stain: Golden Pecan / Early American

